When Fairness Metrics Disagree Under Shift: A Stylized Case Study of a Medical Follow-up Risk Score under Hospital Deployment Shift

When a “fair” hospital model travels and quietly becomes unfair.

Meixuan Chen

Department of Electrical and Computer Engineering

Algorithmic Fairness Seminar – Summer 2026

Motivation: Fairness Metrics in Practice

- Group fairness metrics (e.g., demographic parity, equalized odds) are defined under a fixed data-generating process.
- Real-world deployments are rarely stable—especially when risk scores are moved across hospitals or regions.
- A hospital may monitor one fairness metric (e.g., demographic parity) and implicitly assume other notions of equity move in the same direction.

This project asks (RQ1–RQ3):

1. How do standard group fairness metrics behave under controlled shifts, even when the underlying classifier remains fixed?
2. When demographic parity and error-based metrics disagree, does the discrepancy reflect degradation in clean outcomes or measurement distortion induced by recorded-label noise?
3. How do the three mitigation strategies change fairness metrics and utility?

Method: Distribution Shift Mechanisms

1. **Group-conditioned base-rate shift:** Varies group-specific prevalence.
   - $\Pr(Y = 1 \mid S = 0) = 0.3 + 0.2\alpha$, $\Pr(Y = 1 \mid S = 1) = 0.3 - 0.2\alpha$, $\alpha \in [0, 1]$.
   - Mimics different disease prevalence across groups in the target hospital.
2. **Group-specific covariate shift:** Changes feature scale for one group.
   - Multiplicative factor $\gamma \in [1, 4]$ on informative feature X_1.
   - Preserves $P(Y \mid X, S)$; mimics case-mix or feature-distribution differences.
3. **Group-specific label noise shift:** Flips labels asymmetrically.
   - Positive labels in group A flipped with prob. $\beta \in [0, 0.3]$.
   - Negative labels in group B flipped with prob. $\beta/2$.
   - Mimics asymmetric diagnostic coding errors.

Method: Fairness and Calibration Metrics

Demographic Parity Difference (DP diff)

$$\left|\Pr(\hat{Y}=1 \mid S=0) - \Pr(\hat{Y}=1 \mid S=1)\right|$$

Measures balance in positive prediction rates across groups.

Equalized Odds Difference (EO diff)

$$\max\left(|\mathrm{TPR}_0 - \mathrm{TPR}_1|, |\mathrm{FPR}_0 - \mathrm{FPR}_1|\right)$$

Worst-case discrepancy in both true and false positive rates.

True-Positive-Rate Gap (TPR gap)

$$|\mathrm{TPR}_0 - \mathrm{TPR}_1|$$

Equal-opportunity violation relative to the chosen outcome definition.

Also track Expected Calibration Error (ECE) globally and per-group; report ECE gap.

Method: Mitigation Strategies

Three interventions, each representing a distinct target-data access regime:

1. **KDE Importance Reweighting:**
   - Estimates density ratio $w(x) = \hat{p}_{\text{target}}(x)/\hat{p}_{\text{train}}(x)$.
   - Reweights logistic loss; justified under pure covariate shift.
2. **Post-hoc Threshold Adjustment (Equal Opportunity):**
   - Uses labeled target calibration set $(X, \tilde{Y}, S)$.
   - Matches the larger group TPR observed at threshold 0.5.
3. **Target-Distribution Retraining:**
   - Retrains same model on labeled target sample $(X, \tilde{Y})$.
   - Supervised reference, not a fairness-specific fix.

Experimental Setup

Data and Models:

- Source training, shifted evaluation, target retraining: $N = 5000$ each.
- KDE adaptation and threshold calibration: $N = 1000$ each.
- Classifier: correctly specified logistic regression on $X = (X_1, X_2)$ without S (max_iter $= 1000$, default L2, $C = 1.0$).
- Fairness metrics via `fairlearn.metrics`; ECE with 10 bins.
- Results averaged over 3 seeds for one-dimensional sweeps, 5 seeds for the danger-zone grid; bands show ± 1 standard deviation.

Shift Parameter Grids:

- Base-rate shift: $\alpha \in [0, 1]$, 11 steps.
- Covariate shift: $\gamma \in [1, 4]$, 10 steps.
- Label noise: $\beta \in [0, 0.3]$, 10 steps.
- Joint (α, β) grid: $11 \times 10 = 110$ cells.

Results: One-Dimensional Metric Behaviour

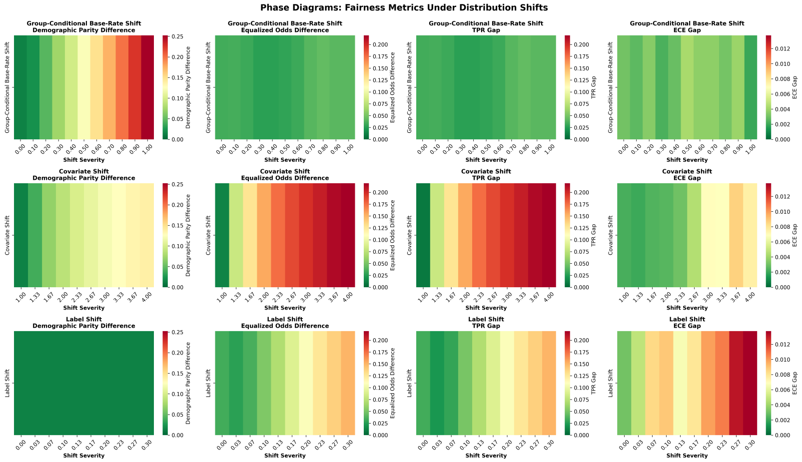

Figure S2: Three-seed mean metrics versus shift severity.

Results: Key Empirical Findings (Part 1)

Group-conditioned base-rate shift:

- DP increases from 0.014 to 0.252 as α goes from 0 to 1.
- TPR gap remains between 0.027 and 0.043.
- Metrics react very differently to the same prevalence change.

Pure covariate shift:

- DP reaches 0.139, EO reaches 0.219, TPR gap reaches 0.219 at $\gamma = 4$.
- ECE gap remains below 0.009—little between-group calibration disparity.
- Metrics worsen but at different rates; they do not signal identical concern.

Asymmetric label noise:

- DP stays flat at 0.014.
- Recorded-label TPR gap worsens from 0.034 to 0.150.
- Clean-outcome TPR gap remains 0.034—the apparent degradation is measurement distortion.

Results: The Danger Zone (Joint Prior + Label Noise)

Define relative to the no-shift cell:

$$\Delta_D = \mathrm{DP}(\alpha,\beta) - \mathrm{DP}(0,0), \quad \Delta_T = \mathrm{TPRGap}_{\tilde{Y}}(\alpha,\beta) - \mathrm{TPRGap}_{\tilde{Y}}(0,0).$$

The *metric-monitoring danger zone* rule is

$$\Delta_T \geq 0.05 \ \wedge \ \left[\Delta_D \leq 0.02 \ \vee \ \Delta_T \geq 2\max(\Delta_D, 0)\right].$$

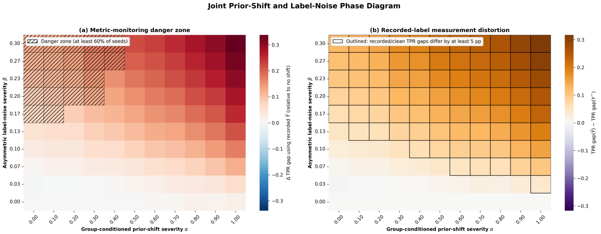

Figure 1: Hatched cells satisfy the rule in at least 3 of 5 seeds; panel (b) compares recorded and clean TPR gaps.

- 20 of 110 cells are dangerous, beginning near $\beta = 0.17$ at low α.
- 78 cells differ by at least five points between clean and recorded TPR gaps.
- Sensitivity variants retain 12–31 danger cells.

Results: Mitigation Comparison

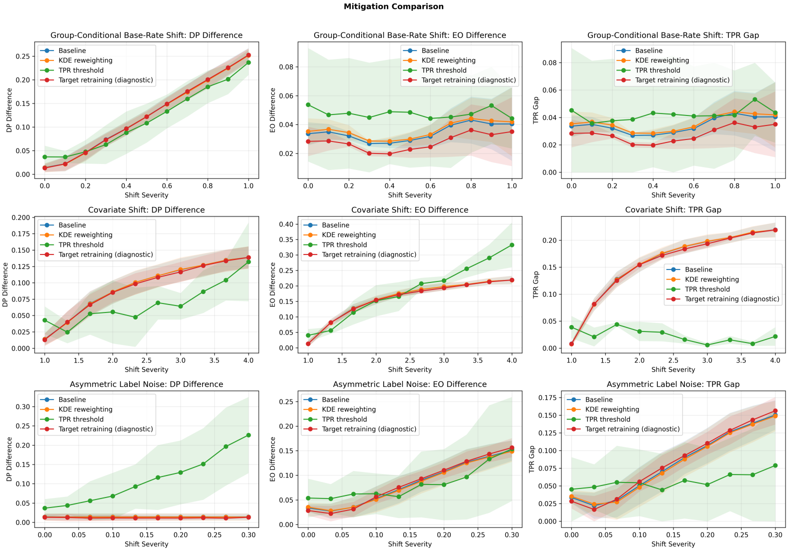

Figure S3: Three-seed mitigation comparison across shifts.

Results: Target Retraining

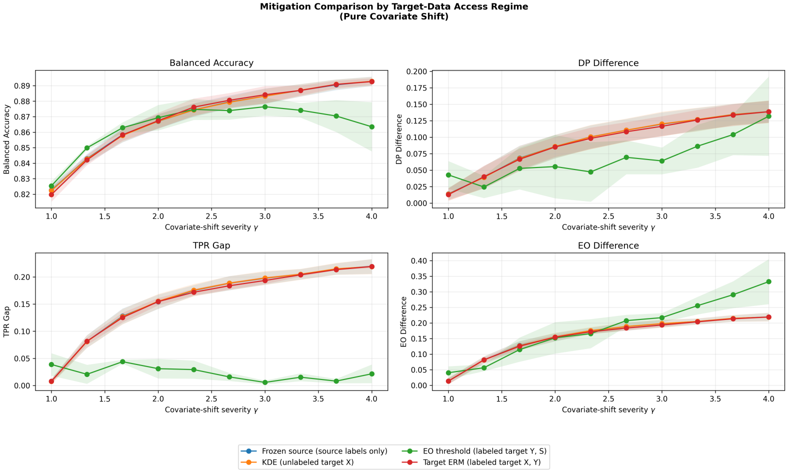

Figure S1: Mitigation under pure covariate shift, organized by target-data access.

Results: Key Empirical Findings (Part 2)—Mitigation

- **KDE Reweighting:**
  - Under pure covariate shift, nearly coincides with the correctly specified frozen model (as expected).
  - In other shift types, it serves as a negative control and does not improve fairness.
- **Threshold Adjustment (Equal Opportunity):**
  - Under covariate shift: reduces max TPR gap from 0.219 to 0.044, but max EO rises to 0.333 (FPR gap grows).
  - At $\gamma = 4$, balanced accuracy drops from 0.893 to 0.863.
  - Under label noise: raises DP from 0.014 to 0.178 (at $\beta = 0.3$).
  - Explicitly reducing TPR gap can increase DP or FPR-driven EO.
- **Target ERM:**
  - Remains close to frozen model under pure covariate shift; not a universal fairness fix.

Discussion: Implications for Practice

1. A stable DP value can mask large changes in error-based fairness.
   - Asymmetric label noise is the clearest example.
   - Dashboards monitoring only DP may miss or mischaracterize recorded-outcome disparities.
2. Error-based metrics can change sharply even when predictions are unchanged.
   - Under label noise, this can reflect measurement distortion rather than worse clean-outcome performance.
3. Conclusion depends on both metric and outcome definition.
   - Recorded vs. clean outcomes can produce entirely different conclusions.
   - Data quality and documentation noise matter for fairness monitoring.

Recommendation

Track at least DP (referral volume), TPR gaps (missed necessary follow-up), and calibration gaps (how equally clinicians can trust scores) side by side. Also audit outcome definitions for recording bias.

Conclusion and Future Work

Key Takeaways:

- Stable DP can be misleading; error-based metrics and outcome definitions are essential.
- Mitigations have trade-offs and respect different data-access assumptions.
- Caution against over-interpreting any single fairness statistic as a stable "fairness guarantee" after deployment shifts.

Future Directions:

1. Extend to richer feature spaces, multi-class, and causal outcome definitions.
2. Wider uncertainty quantification and model families.
3. Explore randomized equalized-odds post-processing.

References

1. Asiaee, A., & Aryan, K. (2026). Fairness under group-conditional prior probability shift: Invariance, drift, and target-aware post-processing. *arXiv:2602.05144*.
2. Chen, Y., Raab, R., Wang, J., & Liu, Y. (2022). Fairness transferability subject to bounded distribution shift. *arXiv:2206.00129*.
3. Barrainkua, A., Gordaliza, P., Lozano, J. A., & Quadrianto, N. (2022). A survey on preserving fairness guarantees in changing environments. *arXiv:2211.07530*.
4. Shao, M., Li, D., Zhao, C., Wu, X., Lin, Y., & Tian, Q. (2024). Supervised algorithmic fairness in distribution shifts: A survey. *arXiv:2402.01327*.